

The Density-Two Hexagonal h_2 System Has Lower Frame Bound Zero

Counterexample packet for arXiv:1710.00753

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Status: candidate full negative answer, likely valid; human review requested. The Gabor system generated by the second Hermite function and the hexagonal lattice of density 2 does <i>not</i> have a positive lower frame bound. Its integer-density Janssen symbol vanishes exactly at a three-torsion point. The cancellation follows from the cubic theta product and a weight-four identity on $\Gamma_0(3)$, not from numerical approximation.
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1 The source question

Faulhuber [1] asks whether the Gabor system generated by the second Hermite function h_2 and the hexagonal lattice of density 2 has a positive lower frame bound. The paper reports numerical evidence for a frame with lower bound approximately 0.29.

$L^2(\mathbb{R}^d)$ and an index set $\Lambda \subset \mathbb{R}^{2d}$, we denote the resulting Gabor system by $\mathcal{G}(g, \Lambda)$. The (full) frame set of the window g is given by

$$\mathfrak{F}_{full}(g) = \{\Lambda \subset \mathbb{R}^{2d}, \Lambda \text{ a lattice} \mid \mathcal{G}(g, \Lambda) \text{ is a frame}\}.$$

Inspired by the work of Lemvig [15], the original intention of this short note was to show up simple restrictions for the full frame set of odd (1-dimensional) Hermite functions by showing that certain sums vanish, but the restriction to this very special class of functions turned out to be unnecessary. Unfortunately, we do not get any new insights into the frame set of even (Hermite) functions. Among other counterexamples, Lemvig showed that the square lattice of density 2 does not generate a Gabor frame for the second Hermite function (the Gaussian being indexed as 0-th Hermite function), which was the first known obstruction to the frame set of the second Hermite function. Numerical inspections suggest that, for the second Hermite function, among all separable lattices of density 2 the square lattice is the only lattice which does not yield a Gabor frame, in particular, in case of the square lattice the lower frame bound is zero and it yields the global minimum of the lower frame bound seen as a function of the lattice parameters. This example stands in sharp contrast to the results given in [3], where it is shown that under the same assumptions, but using the Gaussian instead of the second Hermite function, the square lattice gives the global maximum of the lower frame bound seen as a function of the lattice parameters. The common theme, however, is that in both cases the highest possible symmetry of the lattice leads to extremal frame bounds. It was proven in [2] that, for a Gabor frame of even redundancy with standard Gaussian window, the hexagonal lattice yields the smallest upper frame bound among all lattices. We conjecture that the hexagonal lattice should also give the largest lower frame bound in this case. So, we pose the following question: For the second Hermite function, does the Gabor system generated by the hexagonal lattice of density 2 have a positive lower frame bound? The results by Lemvig tempt us to think that this might not be the case, but numerical inspections say that we actually have a Gabor frame with approximate lower frame bound 0.29...

Our main result, however, concerns odd windows in Feichtinger's algebra which we denote by $S_0(\mathbb{R}^d)$ (another common notation is $M^1(\mathbb{R}^d)$).

Theorem 1.1 (Main Result). *Let $g \in S_0(\mathbb{R}^d)$ be an odd function, i.e., $g(t) = -g(-t)$ and let $\Lambda \subset \mathbb{R}^{2d}$ be a symplectic lattice in the time-frequency plane. If $\text{vol}(\Lambda) = 2^{-d}$ then $\mathcal{G}(g, \Lambda)$ cannot be a Gabor frame. or. in shorter notation:*

Figure 1: The source question and its numerical prediction (PDF page 2).

The source also proves the exact tool needed here. At even integer density, the sharp lower frame bound is the essential infimum of the absolutely convergent Janssen symbol

$$\phi(z) = \text{vol}(\Lambda)^{-1} \sum_{\lambda^\circ \in \Lambda^\circ} \mathcal{A}g(\lambda^\circ) e^{2\pi i \sigma(\lambda^\circ, z)}, \quad (1)$$

where Λ° is the adjoint lattice and $\mathcal{A}g$ is the ambiguity function. Thus one exact zero of ϕ rules out a frame.

2 Proof intuition

At density 2, the adjoint of a normalized hexagonal lattice is its dilate by 2. At a natural three-torsion point of the torus, the characters in (1) become the cubic roots ω^{k-l} . Since the ambiguity function of h_2 is a Laguerre polynomial times a Gaussian, the symbol becomes a second differential expression in the cubic theta series

$$\sum_{k,l} \omega^{k-l} e^{-u(k^2+kl+l^2)}.$$

This series is the eta quotient $\eta(\tau)^3/\eta(3\tau)$. The density fixes $\tau = i/\sqrt{3}$, the Fricke-symmetric point of level 3. The Ramanujan derivative equation reduces the differential expression to $5\mathcal{E}^2 - E_4$,

where $\mathcal{E}$ is the normalized weight-two Eisenstein series on $\Gamma_0(3)$. A one-coefficient Sturm-bound identity gives $E_4 = 5\mathcal{E}^2$ at that point, forcing the symbol to vanish.

3 The hexagonal Janssen symbol

Put

$$H = \sqrt{\frac{2}{\sqrt{3}}} \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}, \quad \Lambda = 2^{-1/2} H \mathbb{Z}^2. \quad (2)$$

Then $\det H = 1$, so $\text{vol}(\Lambda) = 1/2$, and in dimension one H is symplectic. Consequently

$$\Lambda^\circ = \sqrt{2} H \mathbb{Z}^2 = 2\Lambda. \quad (3)$$

Every density-two hexagonal lattice is obtained from this representative by a phase-space rotation or reflection; the Hermite ambiguity function is radial, so the frame bounds are unchanged.

For the standard normalized second Hermite function, direct Gaussian integration gives

$$\mathcal{A}h_2(x, \xi) = e^{-\pi(x^2 + \xi^2)/2} L_2(\pi(x^2 + \xi^2)), \quad L_2(t) = 1 - 2t + \frac{t^2}{2}. \quad (4)$$

Let

$$z_0 = 2^{-1/2} H \begin{pmatrix} 1/3 \\ 1/3 \end{pmatrix}, \quad \omega = e^{2\pi i/3}.$$

For $\lambda^\circ = \sqrt{2} H(k, l)^T$, symplecticity and (2) give

$$\sigma(\lambda^\circ, z_0) = \frac{k-l}{3}, \quad |\lambda^\circ|^2 = \frac{4}{\sqrt{3}}(k^2 + kl + l^2). \quad (5)$$

Set $Q(k, l) = k^2 + kl + l^2$ and $u_0 = 2\pi/\sqrt{3}$. Substituting (4)–(5) into (1) yields

$$\frac{\phi(z_0)}{2} = \sum_{k, l \in \mathbb{Z}} \omega^{k-l} e^{-u_0 Q(k, l)} (1 - 4u_0 Q(k, l) + 2u_0^2 Q(k, l)^2). \quad (6)$$

4 Exact cubic-theta cancellation

For $u > 0$, define

$$B(u) = \sum_{k, l \in \mathbb{Z}} \omega^{k-l} e^{-u Q(k, l)}.$$

The series and all its derivatives converge absolutely. Equation (6) is

$$\frac{\phi(z_0)}{2} = B(u_0) + 4u_0 B'(u_0) + 2u_0^2 B''(u_0). \quad (7)$$

Let $q = e^{2\pi i \tau} = e^{-u}$. The classical Borwein cubic theta product [2] is

$$B(u) = b(\tau) := \sum_{k, l \in \mathbb{Z}} \omega^{k-l} q^{Q(k, l)} = \frac{\eta(\tau)^3}{\eta(3\tau)}. \quad (8)$$

We need one elementary level-three modular identity. Write E_2, E_4 for the normalized Eisenstein series and set

$$\mathcal{E}(\tau) = \frac{3E_2(3\tau) - E_2(\tau)}{2}.$$

Then

$$\mathcal{E}(\tau)^2 = \frac{E_4(\tau) + 9E_4(3\tau)}{10}. \quad (9)$$

Indeed, both sides are weight-four modular forms on $\Gamma_0(3)$. The Sturm bound is $\lfloor (4/12)[\text{SL}_2(\mathbb{Z}) : \Gamma_0(3)] \rfloor = 1$, and their expansions both begin $1 + 24q + O(q^2)$.

Now take $\tau_0 = i/\sqrt{3}$, so $u_0 = 2\pi/\sqrt{3}$ and $3\tau_0 = -1/\tau_0$. The modular transformations of E_2, E_4 give, with $e = E_2(\tau_0)$ and $f = E_4(\tau_0)$,

$$E_2(3\tau_0) = -\frac{e}{3} + \frac{2\sqrt{3}}{\pi}, \quad E_4(3\tau_0) = \frac{f}{9}. \quad (10)$$

It follows immediately from (9) that

$$\mathcal{E}(\tau_0)^2 = \frac{f}{5}. \quad (11)$$

For completeness, let $\delta = q \, d/dq = -d/du$ and

$$a(\tau) = \delta \log b(\tau) = \frac{E_2(\tau) - E_2(3\tau)}{8}. \quad (12)$$

Hence $B' = -aB$ and $B'' = (a^2 + \delta a)B$. Ramanujan's identity $\delta E_2 = (E_2^2 - E_4)/12$, remembering the factor 3 when differentiating $E_2(3\tau)$, gives at τ_0

$$\delta a = \frac{1}{96} \left[e^2 - f - 3 \left(E_2(3\tau_0)^2 - \frac{f}{9} \right) \right]. \quad (13)$$

Substitution of (10), (12), and (13) into (7) simplifies exactly to

$$\frac{B + 4u_0 B' + 2u_0^2 B''}{B} = \frac{\pi^2}{54} (5\mathcal{E}(\tau_0)^2 - f) = 0, \quad (14)$$

where the last equality is (11). Since η has no zeros in the upper half-plane, $B(u_0) \neq 0$, so the division in (14) is legitimate.

Theorem 1. *The Gabor system $\mathcal{G}(h_2, \Lambda)$, with Λ the hexagonal lattice of density 2, is not a frame. Its sharp lower frame bound is zero.*

Proof. Equations (7) and (14) give $\phi(z_0) = 0$. The Janssen series is continuous because h_2 is Schwartz, and the source's sharp-bound formula (1) says its essential infimum is the lower frame bound. A continuous symbol with a zero cannot have positive essential infimum. Hence the lower frame bound is zero and the system is not a frame. $\square$

5 Verification, scope, and novelty

The accompanying script evaluates the truncated sum in (6) at 100-digit precision: square cutoffs 2, 3, 4, 5, 6, 8, 10 converge successively from $4.45 \cdot 10^{-8}$ to below 10^{-99} . It also checks symbolically the reduction in (14). These computations audit signs and normalizations; the proof is the exact modular calculation above.

The result completely answers the source's displayed question negatively and contradicts only its numerical prediction, not its theorem about odd windows. On 12 August 2026, the four run indexes and bounded exact/current searches for the source identifier, exact question, claimed 0.29 bound, hexagonal/nonseparable h_2 systems, and later citations found no explicit resolution. Later separable-lattice frame-set results do not cover this question. Novelty confidence is moderate-high.

Human-review recommendation. Review as a full negative answer. The most important checks are the lattice normalization (2)–(5), the cubic theta product (8), and the factor-of-three derivative in (13).

References

- [1] M. Faulhuber, *A Short Note on the Frame Set of Odd Functions*, Bull. Aust. Math. Soc. 98 (2018), 481–493; arXiv:1710.00753.
- [2] J. M. Borwein and P. B. Borwein, *A Cubic Counterpart of Jacobi’s Identity and the AGM*, Trans. Amer. Math. Soc. 323 (1991), no. 2, 691–701.